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Quantum computing applications, challenges, and prospects in financial portfolio optimization

Omoshola S. Owolabi [1,*], Prince C. Uche [1], Nathaniel T. Adeniken [1], Victoria Tanoh [2] and Oluwabukola G. Emi-Johnson [3]

[1] *Department of Data Science, Carolina University, Winston Salem - North Carolina, USA.*
[2] *Applied Science and Technology Program, NC A&T State University, Greensboro, North Carolina, USA.*
[3] *Department of Statistical Sciences, Wake Forest University, North Carolina, USA.*

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Abstract

Quantum computing has the potential to improve financial portfolio optimization by addressing scalability and computational complexity issues. This article explores the application of quantum algorithms to portfolio optimization. It begins by discussing the limitations of classical optimization methods and introduces the basics of quantum computing. Two key quantum algorithms, quantum annealing and the Quantum Approximate Optimization Algorithm (QAOA), are presented in detail. These algorithms are applied to solve the Quadratic Unconstrained Binary Optimization (QUBO) formulation of the portfolio optimization problem. The article provides a high-level quantum algorithm, along with its pseudo-code Python implementation. The potential computational speedup of quantum algorithms is analyzed, highlighting the theoretical quadratic speedup over classical methods. However, the article also acknowledges the challenges and limitations currently facing quantum computing. It also concludes by emphasizing the promising future of quantum computing in finance and encourages further research to unlock the full potential of quantum technologies in portfolio optimization and other complex financial problems.

Keywords: Quantum; Portfolio; Optimization; Annealing; Binary; Computational Complexity; Algorithms

1. Introduction

Portfolio optimization is a critical task in finance that aims to allocate assets in a way that maximizes returns while minimizing risk. The seminal work of Harry Markowitz [1] on mean-variance optimization laid the foundation for modern portfolio theory. However, as the number of assets grows, the computational complexity of portfolio optimization increases exponentially, making it challenging to solve using classical optimization methods.

Quantum computing has emerged as a promising approach to tackle computationally intensive problems across various domains, including finance. By harnessing the principles of quantum mechanics, such as superposition and entanglement, quantum computers can explore vast solution spaces efficiently. This has led to the development of quantum algorithms that have the potential to provide significant speedups over their classical counterparts.

In recent years, there has been growing interest in applying quantum computing to financial problems, particularly portfolio optimization. Quantum algorithms, such as quantum annealing and the Quantum Approximate Optimization Algorithm (QAOA), have shown promise in solving optimization problems. These algorithms can be adapted to tackle the Quadratic Unconstrained Binary Optimization (QUBO) formulation of the portfolio optimization problem.

* Corresponding author: Omoshola S. Owolabi

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However, the field of quantum computing is still in its early stages, and there are several challenges that need to be addressed before quantum computers can be widely adopted for financial applications. These challenges include the limited number of qubits available in current quantum hardware, the presence of noise and decoherence, and the difficulty of integrating quantum systems with classical financial infrastructure.

2. Fundamentals of Quantum Computing

Before the exploration of the application of quantum computing to portfolio optimization, it is essential to understand the basic concepts of quantum computing, including qubits, quantum gates, and quantum algorithms.

2.1. Qubits

The fundamental unit of quantum information is the quantum bit, or qubit. Unlike classical bits, which can only be in one of two states (0 or 1), a qubit can be in a superposition of both states simultaneously. The state of a qubit is represented by a linear combination of the basis states $|0\rangle$ and $|1\rangle$:

$$[|\psi\rangle = \alpha|0\rangle + \beta|1\rangle] \quad (1)$$

where α and β are complex numbers called amplitudes, satisfying *|α|^2 + |β|^2 = 1.*

The probabilities of measuring the qubit in the *|0⟩ or |1⟩* state are given by *|α|^2 and |β|^2*, respectively.

2.2. Quantum Gates

Quantum gates are the basic operations that can be applied to qubits. They are represented by unitary matrices, which preserve the norm of the quantum state. Some common single-qubit gates include:

- Pauli-X gate (X): Equivalent to the classical NOT gate, it flips the state of the qubit.
- Pauli-Y gate (Y): Rotates the qubit by π radians around the Y-axis of the Bloch sphere.
- Pauli-Z gate (Z): Rotates the qubit by π radians around the Z-axis of the Bloch sphere.
- Hadamard gate (H): Creates an equal superposition of the basis states.

Multiple qubits can be entangled using two-qubit gates, such as the controlled-NOT (CNOT) gate, which flips the state of the target qubit conditional on the state of the control qubit.

2.3. Quantum Algorithms

Quantum algorithms leverage the properties of qubits and quantum gates to perform computations that are infeasible or inefficient on classical computers. Some well-known quantum algorithms include:

- Shor's algorithm: Factors large integers exponentially faster than the best-known classical algorithm.
- Grover's algorithm: Provides a quadratic speedup for unstructured search problems.
- Quantum Fourier Transform (QFT): A key component in many quantum algorithms, including Shor's algorithm and quantum phase estimation.

In the context of optimization problems, two prominent quantum algorithms are:

i. Quantum Annealing: Utilizes the principles of adiabatic quantum computation to find the ground state of a given Hamiltonian, which corresponds to the optimal solution of the optimization problem.
ii. Quantum Approximate Optimization Algorithm (QAOA): A hybrid quantum-classical algorithm that alternates between applying cost and mixing Hamiltonians to prepare a quantum state that encodes the solution to the optimization problem.

These quantum algorithms, along with others, have the potential to provide significant speedups over classical methods for certain classes of problems, including portfolio optimization.

Mathematically, quantum computing exploits the principles of quantum mechanics, such as superposition and entanglement, to perform computations. The fundamental unit of quantum information is the qubit, which can be in a superposition of the basis ($|0\rangle$)and ($|1\rangle$):

$$[|\psi\rangle = \alpha|0\rangle + \beta|1\rangle] \tag{1}$$

where (α) and (β) are complex amplitudes satisfying. $(|\alpha|^2 + |\beta|^2 = 1)$ Multiple qubits can be combined using the tensor product $\otimes$ to form a multi-qubit system:

$$[\,|\psi_1\rangle \otimes |\psi_2\rangle = (\alpha_1|0\rangle + \beta_1|1\rangle) \otimes (\alpha_2|0\rangle + \beta_2|1\rangle)\,] \tag{1}$$

Quantum operations are performed using unitary matrices (U), which satisfy

$(U^\dagger U = UU^\dagger = I)$, where $(U^\dagger)$ is the conjugate transpose of (U) and (I) is the identity matrix.

3. Portfolio Optimization Formulation

The classical portfolio optimization problem, as formulated by Markowitz [1], involves minimizing the portfolio variance while achieving a target return. The problem can be expressed as a quadratic programming problem:

$$\begin{gathered}\text{Minimize } \tfrac{1}{2}\mathbf{w}^{\mathbf{T}}\Sigma\mathbf{w} \\ \text{subject to } \mu^T\mathbf{w} = \mu_p \\ \mathbf{e}^{\mathbf{T}}\mathbf{w} = 1 \\ 0 \le w_i \le 1 \quad \forall i\end{gathered} \tag{1}$$

Where (w) is the vector of asset weights, (Σ) is the covariance matrix, (μ) is the vector of expected returns, (μ_p) is the desired portfolio return, and (e) is a vector of ones.

The Lagrangian formulation of the problem introduces Lagrange multipliers (λ_1) and (λ_2):

$$\mathcal{L}(\mathbf{w}, \lambda_1, \lambda_2) = \frac{1}{2}\mathbf{w}^{\mathbf{T}}\Sigma\mathbf{w} - \lambda_1(\mu^T\mathbf{w} - \mu_p) - \lambda_2(\mathbf{e}^{\mathbf{T}}\mathbf{w} - 1) \tag{2}$$

The optimal solution satisfies the Karush-Kuhn-Tucker (KKT) conditions [2]:

$$\begin{gathered}\nabla\mathcal{L}(\mathbf{w}^*, \lambda_1^*, \lambda_2^*) = 0 \\ \mu^T\mathbf{w}^* = \mu_p \\ \mathbf{e}^{\mathbf{T}}\mathbf{w}^* = 1 \\ \lambda_1^* \ge 0, \lambda_2^* \ge 0\end{gathered} \tag{2}$$

4. Quantum Algorithms for Optimization

4.1. Quantum Annealing

Quantum Annealing solves optimization problems by evolving a quantum system towards its ground state [3]. The system is governed by the time-dependent Hamiltonian:

$$H(t) = A(t)H_B + B(t)H_P \tag{3}$$

where (H_B) is the initial Hamiltonian, representing an easy-to-prepare ground state, and (H_P) is the problem Hamiltonian encoding the optimization problem. The functions $(A(t))$ and $(B(t))$ control the annealing schedule, with$(A(0) \gg B(0))and(A(T) \ll B(T))$, where (T) is the total annealing time.

The adiabatic theorem [4] states that if the annealing process is performed slowly enough, the system will remain in its ground state throughout the evolution, leading to the optimal solution of the problem Hamiltonian (H_P).

For portfolio optimization, the problem Hamiltonian can be constructed as:

$$H_P = \sum_{i,j} Q_{ij} Z_i \otimes Z_j \ (3) \tag{4}$$

where (Q_{ij}) are the coefficients derived from the quadratic form of the objective function, and (Z_i) are Pauli-Z operators acting on the $(i) - th$ qubit.

4.2. Quantum Approximate Optimization Algorithm (QAOA)

QAOA is a hybrid quantum-classical algorithm for combinatorial optimization problems [5]. The algorithm involves preparing a parameterized quantum state:

$$|\psi(\vec{\gamma}, \vec{\beta})\rangle = U_B(\beta_p) U_C(\gamma_p) \cdots U_B(\beta_1) U_C(\gamma_1) |s\rangle \tag{5}$$

where $(U_C(\gamma) = \exp(-i\gamma H_C)) and (U_B(\beta) = \exp(-i\beta H_B))$ are unitary operators corresponding to the cost and mixing Hamiltonians, respectively. The initial state $(|s\rangle)$ is a superposition of all possible solutions.

The cost Hamiltonian (H_C) encodes the objective function, while the mixing Hamiltonian (H_B) introduces transitions between different solutions. The optimal parameters $(\overrightarrow{\gamma^*})$and $(\overrightarrow{\beta^*})$are found by minimizing the expectation value of the cost Hamiltonian:

$$\min_{\vec{\gamma}, \vec{\beta}} \langle \psi(\vec{\gamma}, \vec{\beta}) | H_C | \psi(\vec{\gamma}, \vec{\beta}) \rangle \tag{5}$$

4.3. Quantum Algorithm for Portfolio Optimization

Algorithm 1: Quantum Algorithm for Portfolio Optimization

The application of quantum algorithms for portfolio optimization, as proposed by Doe [12], leverages the principles of quantum annealing to efficiently solve the problem. By encoding the problem into a QUBO formulation and constructing the corresponding cost Hamiltonian, the algorithm exploits quantum effects to explore the solution space.

Algorithm 1: Quantum Algorithm for Portfolio Optimization [12]

Input:

- Set of assets $A = \{a_1, a_2, \ldots, a_n\}$
- Expected returns $\mu \in \mathbb{R}^n$
- Covariance matrix $\Sigma \in \mathbb{R}^{n \times n}$
- Target portfolio return μ_p
- Risk aversion parameter λ

Output:

- Optimal portfolio allocation $x^* \in \{0, 1\}^n$

Procedure:

1. Encoding the portfolio optimization problem as a Quadratic Unconstrained Binary Optimization (QUBO) problem: minimize $_x x^T Q\, x$
   where:
   i. $Q \in \mathbb{R}^{n \times n}$ is the QUBO matrix,
   ii. and $x \in \{0, 1\}^n$ is the binary decision vector representing the asset allocation.

2. Constructing the cost Hamiltonian H_C based on the QUBO formulation: $H_C = \sum_{I,j} Q_{ij} \sigma_i^z \sigma_j^z$
   where σ_i^z is the Pauli-Z operator acting on the $_i - th$ qubit.

3. Defining the mixing Hamiltonian H_B to introduce transitions between different states: $H_B = \sum_i \sigma_i^x$where $\sigma_i^x$is the Pauli-X operator acting on the $_i - th$ qubit.

4. Initializing the quantum system in the equal superposition state: $|s\rangle = \left(1/\sqrt{2^n}\right)\sum_x |x\rangle$

5. Applying the quantum annealing evolution: $U(T) = T\exp(-i\int_0^T (A(t)H_B + B(t)H_C)dt)$
   where
   i. T is the time-ordering operator,
   ii. and $A(t)$ and $B(t)$ are the annealing schedule functions satisfying $A(0) \gg B(0)$ ad $A(T) \ll B(T)$.

6. Measuring the final state in the computational basis to obtain a candidate solution x'.

7. Repeating steps 4-6 for a fixed number of iterations or until a satisfactory solution is found.

8. Selecting the best portfolio allocation x^* that minimizes the objective function while satisfying the constraints: $x^* = \mathrm{argmin}'_x x'^T Q x' : \mu^T x' \geq \mu_p, \sum_i x'_i = 1$

9. Return the optimal portfolio allocation x^*.

The quantum algorithm presented above provides a high-level overview of the steps involved in solving the portfolio optimization problem using quantum annealing. To better understand the implementation details, let's consider the pseudo code for this algorithm.

Algorithm 1: Python Implementation pseudo code

This algorithm requires a quantum computing framework like Qiskit, Cirq, or QuTiP, as well as access to a quantum computer or simulator.

This is a simplified version using the IBM framework.

Algorithm 1 Quantum Portfolio Optimization

```
procedure ENCODE_QUBO(μ, Σ, μ_p, λ)
    n ← len(μ)
    Q ← np.zeros((n, n))
    for i ← 0 to n − 1 do
        for j ← 0 to n − 1 do
            Q[i][j] ← Σ[i][j] − λ · μ[i] · μ[j]
        end for
    end for
    return Q
end procedure
procedure CONSTRUCT_COST_HAMILTONIAN(Q)
    n ← len(Q)
    H_C ← Σ_{i=0}^{n−1} Σ_{j=0}^{n−1} Q[i][j] · (Γ σ_i^z σ_j^z)
    return H_C
end procedure
procedure CONSTRUCT_MIXING_HAMILTONIAN(n)
    H_B ← Σ_{i=0}^{n−1} Γ σ_i^x
    return H_B
end procedure
procedure QUANTUM_PORTFOLIO_OPTIMIZATION(μ, Σ, μ_p, λ)
    n ← len(μ)
    Q ← encode_qubo(μ, Σ, μ_p, λ)
    H_C ← construct_cost_hamiltonian(Q)
    H_B ← construct_mixing_hamiltonian(n)
    qc ← QuantumCircuit(n)
    qc.h(range(n))
    qc.append(H_C, range(n))
    qc.append(H_B, range(n))
    qc.measure_all()
    backend ← Aer.get_backend('qasm_simulator')
    job ← backend.run(assemble(qc))
    result ← job.result().get_counts()
    return result
end procedure
```

The pseudo code snippet demonstrates the implementation of the key components of the quantum algorithm for portfolio optimization. It includes functions for encoding the problem into a QUBO formulation, constructing the cost and mixing Hamiltonians, and performing the quantum annealing evolution using the IBM Qiskit framework.

5. Computational Complexity and Quantum Speedup

Portfolio optimization is an NP-hard problem, meaning that its computational complexity grows exponentially with the number of assets [6]. Classical algorithms, such as interior-point methods, have a time complexity of $(O(n^3))$ for a portfolio with (n) assets.

Quantum algorithms, such as quantum annealing and QAOA, have the potential to provide a quadratic speedup over classical methods [7]. The adiabatic algorithm for solving quadratic unconstrained binary optimization (QUBO) problems, which can be used to formulate portfolio optimization, has a time complexity of $(O(\sqrt{n}))$ [8].

However, the actual speedup achieved by quantum algorithms depends on various factors, such as the problem instance, the quality of the quantum hardware, and the efficiency of the classical optimization routines used in hybrid algorithms.

6. Challenges and Limitations

Quantum computing presents a promising avenue for advancing portfolio optimization in finance. However, several challenges and limitations currently hinder its practical applicability:

6.1. Noise and Decoherence

Quantum systems are inherently sensitive to environmental noise, which can lead to errors in the computation. The presence of noise can cause the quantum state to deviate from the desired state, resulting in inaccurate results. Decoherence, the loss of quantum coherence over time, limits the depth of quantum circuits that can be reliably executed [9]. Current quantum hardware has limited coherence times, which restricts the complexity of the algorithms that can be implemented. Mitigating the effects of noise and decoherence is an active area of research, with techniques such as quantum error correction [13] and dynamical decoupling [14] being explored to improve the reliability of quantum computations

6.2. Scalability

The number of qubits available in current quantum computers is limited, which restricts the size of optimization problems that can be solved. To tackle real-world portfolio optimization problems with many assets, a significant increase in the number of qubits is necessary. However, scaling up quantum hardware while maintaining high coherence times is an ongoing challenge [10]. The development of more robust and scalable quantum architectures, such as superconducting qubits [15] and trapped ions [16], is crucial for the practical application of quantum computing to large-scale optimization problems.

6.3. Input/Output Bottleneck

Efficiently encoding classical data into quantum states and extracting the results of quantum computations can be a bottleneck in quantum-classical hybrid algorithms [11]. Portfolio optimization involves processing large amounts of financial data, including asset prices, returns, and risk factors. Encoding this classical data into quantum states requires efficient techniques that minimize the overhead and preserve the required precision. Similarly, measuring and interpreting the results of quantum computations can be challenging, especially when dealing with high-dimensional quantum states. Developing efficient input/output interfaces and data encoding schemes is an active area of research [17] to address this bottleneck.

6.4. Approximation Ratios

While quantum algorithms can provide speedups over classical methods, they often come with approximation ratios that are worse than the best-known classical approximations [12]. The approximation ratio quantifies the quality of the solution obtained by an algorithm compared to the optimal solution. Many quantum optimization algorithms, such as the Quantum Approximate Optimization Algorithm (QAOA) [18], have approximation ratios that are not yet competitive with state-of-the-art classical approximation algorithms. Rigorous analysis of the approximation performance of quantum optimization algorithms is an ongoing research area, aiming to improve the guarantees and understand the limitations of these algorithms.

6.5. Integration with Classical Systems

Integrating quantum computers into existing financial systems and workflows presents another challenge. Portfolio optimization is typically part of a larger investment management process that involves data sourcing, risk modeling, trade execution, and reporting. Seamlessly integrating quantum optimization routines into this process requires robust interfaces and compatibility with classical systems [19]. Developing quantum-classical hybrid architectures that leverage the strengths of both quantum and classical computing is an important step towards practical quantum finance applications.

6.6. Talent and Expertise

The field of quantum computing is still in its early stages, and there is a limited pool of talent with expertise in both quantum algorithms and financial domain knowledge. Developing and implementing quantum algorithms for portfolio optimization requires a multidisciplinary skill set, combining quantum information science, optimization theory, and financial mathematics [20]. Fostering collaboration between quantum experts and financial professionals, as well as investing in quantum education and training programs, is crucial for bridging the talent gap and accelerating the adoption of quantum technologies in finance.

Addressing these challenges and limitations requires sustained research efforts, technological advancements, and collaboration between the quantum computing and finance communities. As progress is made in quantum hardware, error correction, and algorithm design, the practical applicability of quantum computing to portfolio optimization and other financial problems will likely improve. Ongoing research and development in these areas are essential for realizing the full potential of quantum computing in finance.

7. Conclusion

The emergence of quantum computing presents a promising avenue for advancing portfolio optimization in finance. By leveraging the principles of quantum mechanics, quantum algorithms such as quantum annealing and QAOA have the potential to provide significant speedups over classical methods. These algorithms can efficiently explore vast solution spaces and tackle the computational complexity that arises in portfolio optimization problems with large asset pools and numerous constraints.

This article has explored the application of quantum computing to portfolio optimization, starting with the mathematical formulation of the problem and the limitations of classical optimization methods. The fundamentals of quantum computing, including qubits, quantum gates, and quantum algorithms, have been introduced to provide a foundation for understanding the quantum approach to optimization.

The quantum algorithm for portfolio optimization, based on quantum annealing and QAOA, has been presented in detail. The algorithm involves encoding the portfolio optimization problem into a QUBO formulation, constructing the cost and mixing Hamiltonians, and performing the quantum annealing evolution. The pseudo-code and a simplified pseudo which can be implemented in Python using the IBM Qiskit framework have been provided to illustrate the implementation of the algorithm.

The potential computational speedup offered by quantum algorithms has been discussed, highlighting the theoretical quadratic speedup over classical methods. However, the article has also emphasized the practical challenges and limitations currently facing quantum computing in finance. These challenges include the limited number of qubits in current quantum hardware, the presence of noise and decoherence, the input/output bottleneck, and the integration with classical financial systems.

Despite these challenges, the prospects of quantum computing in finance are promising. As quantum hardware continues to improve and quantum algorithms become more refined, the application of quantum computing to portfolio optimization and other complex financial problems will likely become increasingly feasible. The development of error correction techniques, the increase in qubit capacity, and the advances in quantum-classical hybrid algorithms will help mitigate the current limitations.

To fully realize the potential of quantum computing in finance, ongoing research and collaboration between the quantum computing and financial mathematics communities are essential. Further investigations into the theoretical foundations of quantum algorithms, the development of efficient quantum implementations, and the exploration of novel quantum-inspired optimization techniques will be crucial in advancing the field.

The integration of quantum computing into existing financial frameworks and the development of quantum-ready financial models will be necessary to bridge the gap between theory and practice. This will require close collaboration between quantum experts, financial professionals, and regulatory bodies to ensure the safe and effective deployment of quantum technologies in the financial industry.

Quantum computing presents an exciting frontier for portfolio optimization and other financial applications. While challenges remain, the potential benefits of quantum algorithms in terms of computational speedup and improved solution quality are significant. As research progresses and quantum technologies mature, we can expect to see a growing impact of quantum computing on the financial landscape. The continued exploration of this field will not only advance our understanding of quantum computing but also revolutionize the way we approach complex financial problems, ultimately leading to more efficient and effective financial decision-making.

Compliance with ethical standards

Disclosure of conflict of interest

No conflict of interest to be disclosed.

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